

COMBINED REQUESTED REVISIONS

- ☐ Revise the Data Availability statement so it describes only this dataset, and move any additional notes to the Usage Notes. [Editor]
- ☐ Clarify whether "Amazon S3 Bucket (Requester Pays)" implies a cost, and remove any charge — the journal does not permit datasets to have an associated fee. [Editor]
- ☒ ~~Merge the SI file into the main text (e.g., into Methods), remove/replace all mentions of supplementary files, renumber and recaption figures/tables accordingly, and ensure Data Descriptor sections contain no additional results or descriptive statistics. [Editor]~~
- ☐ Add an ethics statement confirming that written, informed consent for participation *and* data sharing was obtained. [Editor; relates to R1 Major 1]
- ☐ Confirm in the ethics statement that consent for participation and data sharing was obtained from a parent/guardian for children (relevant to the pediatric/CHOP participants). [Editor; relates to R1 Major 1]
- ☐ Name the specific IRB/ethics committee that approved the study and the data sharing, with the approval/application number; the named board must match at least one author's institution (no "our ethics board"). [Editor; relates to R1 Major 1]
- ☒ ~~Move the Data Overview section to before the Technical Validation section. [Editor]~~
- ☐ Ensure a CC0 or CC-BY license is visible on the dataset page (subject to the human-data policy). [Editor; R1 Major 1]
- ☐ Add a data citation for the dataset to the reference list using a DOI URL, and cite it wherever the dataset is mentioned — primarily in the first sentence of the Data Record describing where the data are deposited. [Editor]
- ☐ Clarify de-identification procedures and date/time handling for the openly shared iEEG data. [R1 Major 1]
- ☐ Add practical reuse information: a file manifest, dataset version, BIDS-validator output (or documented deviations), reviewer access instructions, and stable code info (commit hash or archived DOI). [R1 Major 2]
- ☐ Include package versions or an environment file for the code used to generate the validation figures. [R1 Minor 12; relates to R1 Major 2]
- ☒ ~~Summarize the center-level (HUP vs. CHOP) difference in rater-consensus agreement in the main Technical Validation section. [R1 Major 3]~~
- ☐ Clarify how majority voting was implemented — the channel-level denominator, handling of missing/hidden/artifact channels, ties, stimulation channels, and three vs. more-than-three annotators — and soften the term "gold-standard" given the inter-rater variability. [R1 Major 4]
- ☒ ~~Shorten the Data Overview and reframe it as a dataset validation / reuse example rather than a hypothesis-driven analysis. [R1 Major 5]~~
- ☐ Explain how the 32 patients / 83 seizures were selected from the broader clinical pool: state the total number of SEEG patients seen in the 2021–2024 window and provide inclusion/exclusion criteria (a flow diagram or table). [R1 Major 6; R2 #2]
- ☐ Clarify whether the released dataset is purely sEEG or broader iEEG, since the Methods mention mixed electrodes. [R1 Minor 1]

- ☐ Make the number of annotators consistent — the abstract says three epileptologists while Data Records says three to five clinicians. [R1 Minor 2]
- ☐ Provide a complete data dictionary for `annotations.tsv` and ensure consistent column names. [R1 Minor 3]
- ☒ ~~State whether seizure-onset pattern labels (LVFA, rhythmic activity, discharges) are included in the released dataset. [R1 Minor 4]~~
- ☐ Reconcile the use of Engel vs. ILAE outcome terminology. [R1 Minor 5; relates to R2 #5]
- ☐ Explain the decimal "outcome" values (e.g., 1.1, 2.1) in the participants table — state explicitly if these are Engel subscores (e.g., Ia = 1.1), distinct from the integer ILAE categories defined in the descriptor. [R2 #5]
- ☐ Add a descriptor for the "follow_up" field and reconcile it with the stated 24-month post-surgery assessment (several entries are < 2). [R2 #5]
- ☐ Reconcile the Engel class I count between the text (6) and Table 1 (11). [R1 Minor 6; R2 #4]
- ☐ Explain the Table 1 asterisk and clarify whether percentages use all patients or only those with available data as the denominator. [R1 Minor 7; R2 #4]
- ☐ Clarify the 17 "Unknown" outcome patients — e.g., whether they did not proceed to surgery or were lost to follow-up. [R2 #3]
- ☒ ~~Specify what the error bars / $\pm$ values represent in each relevant figure. [R1 Minor 8]~~
- ☐ Standardize the terminology for stimulation-induced seizures. [R1 Minor 9]
- ☐ Add a minimal Usage Notes example showing how to match an EDF file to its corresponding row in `annotations.tsv`. [R1 Minor 11]
- ☐ Correct minor typographical errors. [R1 Minor 10]
- ☐ Clarify how annotator identity enters the Figure 3D mixed-effects model (the caption describes only patient ID as a random effect), or correct the caption. [R2 #1]

SPECIFIC EDITOR COMMENTS THAT REQUIRE ACTION

These are specific editorial or data checks for your paper that all require a written response in the “response to comments” file to pass peer review. Please make sure all comments have an answer:

*Please revise the Data Availability statement to only outline the dataset that is meant to be described in this work. Please move any other additional notes to the Usage Notes.

[The data availability statement now is restricted to only describing the data in this work.](#)

*Does the notes 'Amazon S3 Bucket (Requester Pays)' mean that access is not free? Please clarify this in the response file. We do not allow for datasets to have a charge associated with them

The data is hosted on Pennsieve as a public dataset, we have confirmed that there are no costs or charges associated with accessing the data. We have added in further instructions in the usage note to clarify instructions for free access.

* Scientific Data does not encourage the use of supplementary information and does not have page limits. It is generally always easier for readers to read the content in one document, not two, and we do suggest papers are split up to support flow (readers will scroll and pick and choose the sections they need). Please merge the current SI file into the main article, adding content to the relevant manuscript section (e.g. methods). If the content is out of scope for any section, please consider removing it (e.g. Data Descriptors should not contain additional results or descriptive statistics – which applies for the main text and SI file). Please edit/remove any mentions of supplementary files in the article file to align with this merging, renaming figures and tables accordingly, and ensuring their captions are listed in the paper.

We have moved two supplementary figures and the associated text to the main text consistent with the format for data descriptors. We have removed the supplementary patient table because the exact same data is present in the dataset itself as “participants.tsv” and we have removed the exact written SOP that was given to clinicians for annotation since the majority of that information is provided in the manual annotation subsection of our methods. There is no longer any supplementary material.

* Please add details to an ethics statement in the paper confirming that written, informed consent for participation and data sharing was obtained. This may be requested if consent is mentioned, but without explicit permission to share data, or is otherwise ambiguous on what participants agreed to.

We have added increased comprehensive reporting of data sharing consent in the participants subsection of our methods.

* Please add details to an ethics statement in the paper confirming that consent for participation and data sharing was obtained from a parent, guardian (for children), or other directly responsible adult (for other vulnerable groups, where applicable).

We have clarified that all adult patients consented to participate and share data, and adults of all pediatric patients provided consent, and assent from pediatric patients was obtained whenever possible.

* Please add or update the ethics statement to indicate which IRB/ethics committee approved the work and provide the relevant approval or application number (if applicable). Please also indicate that they approved the study/protocol and the sharing of data. This ethics board must be named (do not use “our ethics board” or similar) and the name/institution of the board should match that of at least one of the authors’ institutions. If you are an industrial researcher where no academic/institutional ethics board is available, then ethics approvals may be outsourced to other parties in rare circumstances. Please contact the journal if unsure.

We now clarify these approaches for both institutions. Specifically, at HUP, research was approved by the HUP IRB (protocol # 821778). All patients performed consent to retrospective analysis of their data and data sharing. For the first 14 patients at HUP, stimulation was performed as a research protocol under the same IRB, and all patients provided consent to research stimulation; stimulation in the subsequent HUP patients was performed as a clinical protocol).

At CHOP, research was approved by the CHOP IRB (protocol # ***). Stimulation was performed as a research protocol for all patients at CHOP. For patients <18 years old, consent was obtained from parents or guardians, and assent was obtained from the patient when possible. Adult patients ≥18 years old consented on their own behalf.

*Move the data overview section before the technical validation section

The data overview has been moved to before the technical validation section in the manuscript. We do note that the example analysis demonstrating utility of the dataset does semantically reference the technical validation and we believe the description would flow more clearly if the technical validation precedes the data overview.

* Data need to be shared under a CC0 or CC-BY licence, unless subject to human data policy, or required for use of existing third party data where there are restrictions on re-use. Please contact the journal at scientificdata@nature.com if these circumstances apply. Otherwise, please make sure a CC0 or CC-BY licence is visible on the dataset page. Please see here for more information <https://www.nature.com/sdata/policies/repositories>. Please note CC license should not be used as a method to manage human data protection, so again, please contact us if you have these requirements

We now share data under a CC-BY license.

* Please add a data citation for the dataset to the reference list using these instructions (https://www.nature.com/sdata/submission-guidelines#data_citations - note that a DOI URL should be used, not any other link). Please add the reference number to wherever the dataset is mentioned in the text - the main position should be the first part of the Data Record in a sentence describing where the data has been deposited.

We have added this citation, DOI: ****

Reviewer 1

This manuscript presents an open intracranial EEG dataset with multi-expert annotations of seizure onset time, onset channels, and early spread channels for spontaneous and stimulation-induced seizures. The dataset is potentially valuable for benchmarking seizure detection and localization algorithms, especially because open datasets with spatial expert

annotations remain limited. Overall, the work is suitable for Scientific Data, but several issues need to be addressed before the dataset can be fully interpreted and reused.

Major Comments

1. The dataset includes human intracranial EEG, clinical metadata, electrode localizations, and pediatric participants. Please clarify de-identification, date/time handling, data license, and IRB/consent coverage for open sharing of the human iEEG dataset.

We now clarify these approaches for both institutions. Specifically, at HUP, research was approved by the HUP IRB (protocol # 821778). All patients performed consent to retrospective analysis of their data and data sharing. For the first 14 patients at HUP, stimulation was performed as a research protocol under the same IRB, and all patients provided consent to research stimulation; stimulation in the subsequent HUP patients was performed as a clinical protocol).

At CHOP, research was approved by the CHOP IRB (protocol # ***). Stimulation was performed as a research protocol for all patients at CHOP. For patients <18 years old, consent was obtained from parents or guardians, and assent was obtained from the patient when possible. Adult patients ≥18 years old consented on their own behalf.

For both centers, data was de-identified by removing all names, dates, and any other identifiers from all files. Dates and times were date shifted to re-define the EEG recording start time to be January 1st 2000, preserving durations between events. Data from both centers is shared under a CC-BY license.

2. The manuscript states that the dataset follows iEEG-BIDS and is available on Pennsieve, but users need more practical information to reuse it. Please add a file manifest, dataset version, BIDS-validator output or known deviations, reviewer access instructions, and stable code information such as a commit hash or archived DOI.

We have now added in a BIDS-validator statement into figure caption 2 where we show the file manifest to document compliance along with known deviations along with a .bidsignore file to our dataset for additional metadata.

To assist with user access, we have provided additional documentation in the usage notes section of the manuscript for how to download the dataset and how to link between seizure level metadata and recordings.

Still need: final published dataset version, reviewer access instructions (?), and code used to generate figures

3. The center-level difference in rater-consensus agreement between HUP and CHOP should be summarized in the main Technical Validation section.

Thank you for bringing this to our attention, the center-level difference in rater consensus agreement has been reformatted (turned to boxplots) and included in the main text as part of the technical validation section.

4. Please clarify how majority voting was implemented, including the denominator for channel-level labels, handling of missing/hidden/artifact channels, ties, stimulation channels, and three versus more-than-three annotators. The term “gold-standard” should also be softened, if the data show meaningful inter-rater variability.

We have added a subsection to the methods section of the paper titled “Consensus annotation” to clarify the consensus procedures, pasted below for ease of review:

Consensus calculation

For each seizure, we calculated consensus onset time, channels recruited at UEO, and channels seizing during early seizure propagation. Consensus seizure onset time was calculated as the median seizure onset time across annotators. Channels seizing at both UEO and spread were determined by majority voting. For example, a channel was considered a UEO channel for a given seizure if it appeared in a majority of UEO annotations (2 of 3 annotators, or 3 of 5 annotators). Because seizures were exclusively annotated by an odd number of raters (3 or 5), no ties required adjudication. Multi-expert consensus is the gold standard for seizure annotation established by the FDA²⁶. Alongside the consensus, clinician's individual annotations preserve and quantify the inter-rater variability that the consensus label otherwise conceals, which is most pronounced in harder-to-annotate seizures.

We now clarify that our use of the term “gold-standard” refers to the fact that multi-expert consensus is the reference standard established by the FDA for seizure detection and annotation, which we now cite. When calculating consensus, we now clarify that the denominator for each channel was the number of annotators (3 or 5), and we did not restrict any iEEG channels a-priori from their consideration. We now clarify that the annotation SOP provided to experts instructed them to consider the stimulation channels for possible ictal activity (which may be difficult to appreciate given the presence of artifact on these channels).

5. The algorithm benchmarking section is useful as an example of reuse, but some claims read more like a hypothesis-driven results paper. The Data Overview should be shortened and framed more clearly as a dataset validation or reuse example, rather than a hypothesis-driven analysis.

Thank you for pointing this out, we have since modified our data overview section to comply with Scientific Data's reporting standards. We have cut out unnecessary language, specifically removing forward looking and hypothesis generating commentary about goals for future detection algorithms.

6. Please explain how the 32 patients and 83 seizures were selected from the broader clinical pool. A simple flow diagram or table with inclusion/exclusion criteria, excluded recordings, and reasons for exclusion would help readers understand potential selection bias.

We now clarify our inclusion criteria were to select sequential consenting patients from 2021-2024 with at least one stimulation-induced seizure. Of the 38 who met our criteria, we excluded 6 patients due to signal noise during seizure activity or technical challenges acquiring EEG recordings. Stimulation induced seizures and spontaneous seizures that recreated a patient's typical full seizure semiology and morphology were selected from these patients in order to tune and validate automated seizure annotation algorithms.

Minor Comments

1. Please clarify whether the released dataset is purely sEEG or broader iEEG, since the Methods mention mixed electrodes.

We have removed reference to subdural grids and strips, as all patients were implanted solely with depth electrodes (all underwent sEEG recording).

2. The abstract says three epileptologists annotated each seizure, while the Data Records section says three to five clinicians; please make this consistent.

We have changed the abstract to reflect that there were 3 or 5 annotators for each seizure.

3. Please provide a complete data dictionary for annotations.tsv and ensure consistent column names.

We have corrected the data dictionary in the data records to describe the column names of the fields in the annotations.tsv and that they match the descriptions of the fields in the methods and annotations.json. We exclude "Run" from the description since it is sufficiently described in annotations.json.

4. Please state whether seizure-onset pattern labels such as LVFA, rhythmic activity, and discharges are included in the released dataset, if available.

We have amended the dataset to include this information based on feedback from collaborators and to improve the completeness of the dataset. We have included a binary field describing the

presence of LVFA during seizure onset as determined by clinician review. The annotations.tsv subsection of the data records section now reads as follows:

For all seizures, the “LVFA” field denotes the clinician-determined (EC) presence of low-voltage, fast activity on one or more channels during seizure onset based on visual review.

5. Please reconcile the use of Engel versus ILAE outcome terminology.

We were incorrect in saying that we reported ILAE outcomes. We have corrected our documentation of patient outcomes in the participants section of the methods, the patient table, and the participants.json clarifying that we are reporting last available Engel outcomes.

6. The reported number of Engel I outcomes appears inconsistent between the text and Table 1.

We have corrected the number of Engel 1 outcomes and number of patients with outcome data available

7. Please clarify whether percentages in Table 1 use all patients or only patients with available data as the denominator.

We have added a note in the table 1 caption to clarify that “Percentages are calculated using the number of patients with available data as the denominator.”

8. Please specify what the error bars or $\pm$ values represent in each relevant figure.

We have added a section to the data overview paragraphs in addition to the one present in the technical validation section stating that “All values are reported as mean $\pm$ 95% CI.” We also now have included the following in captions for figures with box plots:

Box plots show 0th, 25th, 50th, 75th, and 100th percentiles excluding outliers based on Tukey’s method.

And have further included the following in captions for figures with regression lines:
Shaded area denotes bootstrapped ($n = 1000$) 95% CI of regression line.

9. Please standardize the terminology for stimulation-induced seizures.

We now define the abbreviation “stim seizures” at first use, and use this in place of the longer “stimulation-induced seizures” throughout the manuscript.

10. Correct minor typographical errors.

We have corrected these.

11. The Usage Notes would benefit from a minimal example showing how to match an EDF file to the corresponding row in `annotations.tsv`.

We have added the following example in the usage notes to help users link seizure level metadata in the annotations.tsv to the seizure EEGs:

Users can link seizures to their metadata by linking the subject name in the “patient” field (HUP250) and integer values in the “approximate_onset” field (271748.81) and “run” field (1) of annotations.tsv with the subject, task and run label of the edf file name respectively: “sub-HUP250_ses-postimplant_task-ictal271748_run-01_ieeg.edf.”

12. Please include package versions or an environment file for the code used to generate the validation figures.

To be done

Reviewer 2

Remarks to the Authors:

Ojemann et al. provide a detailed, open-access SEEG seizure dataset with consensus from multiple experts. With 83 seizures from 32 patients and 3+ experts' annotations, the dataset will be unambiguously useful to epilepsy researchers for automated seizure and channel onset identification. The authors provide sufficient validation to support the robustness of the expert review. An example benchmark test against a seizure detection algorithm is provided.

I have provided minor comments and questions below:

1. Figure 3D caption writes “To test whether annotator identity influenced consensus agreement, we fit mixed-effects models with patients as a random intercept” ... “indicating that annotator identity does not meaningfully alter κ values.” The caption states the goal is to test whether annotator identity influences consensus agreement, but the model description includes only patient ID as a random effect. Could the authors clarify how annotator identity enters the model, or correct the caption if the intended question was about patient-level variance?

Thank you for pointing out the unclear description of our statistical test. We have added additional language to the figure caption to clarify that we are using annotator identity as the fixed effect in the model with patient identity as a random intercept. Because there was negligible variability in the patient intercept, we drop the random effect and use a simple OLS

fixed-effect model with annotator identity as the fixed effect. The updated description is pasted below for ease of review:

To test whether annotator identity influenced consensus agreement, we fit mixed-effects models with patients as a random intercept and annotator identity as a fixed effect. Across both onset and spread, the patient-level variance in the mixed-effects model was extremely small (patient intercept variance ≈ 0.001). Given the negligible between-patient variance, we report the corresponding OLS results, which yielded nearly identical parameter estimates and revealed a nonsignificant fixed effect (onset: $F = 1.81$, $p = 0.13$; spread: $F = 1.11$, $p = 0.35$), indicating that annotator identity does not meaningfully alter values.

2. The authors may wish to briefly state the total number of SEEG patients seen during the 2021–2024 window and summarize inclusion/exclusion criteria, so readers can assess whether the 32-patient cohort represents all eligible cases or a selected subset. This does not need to be a formal flowchart but would aid interpretation of the dataset's scope.

We now clarify our inclusion criteria were to select sequential consenting patients from 2021-2024 with at least one stimulation-induced seizure. Of the 38 who met our criteria, we excluded 6 patients due to signal noise during seizure activity or technical challenges acquiring EEG recordings.

Items 3-5 below concern clinical metadata rather than the core dataset and are offered for completeness rather than as major concerns.

3. Seizure freedom score (Engel outcome) is provided in only 15/32 patients, and only 6 of them achieved seizure freedom. The patient outcome is not directly relevant to whether this dataset is useful for algorithm development purposes, but some additional clarification may help a critical reader evaluate the clinical utility. For example, is there any data regarding the 17 patients with 'Unknown' outcome, such as not proceeding with surgical treatment, or these patients never returned for follow-up?

We greatly appreciate the suggestion and have added additional clinical information. Patients with no surgery performed yet or no follow up information are left as unknown (now n/a in the dataset to maintain proper BIDS compliance) while we have added a "Device" level to the outcome variable to denote patients who were implanted with neurostimulation devices but Engel scores do not apply.

4. Table 1 has an asterisk for "Engel 1 outcome, n (%)*" but the asterisk is without explanation. It likely indicates the 40% is based on the total patients with available outcome, not all patients. Additionally, the number is reported as 11 here, where it was 6 in the text: "Surgical outcomes

were available for 15 patients (47%), with 6 patients (40%) achieving Engel class I (seizure-free) outcomes”

We have removed the asterisk from the table, and added a statement in the table caption clarifying that percentages were calculated from the total number of patients with available data. We have also corrected the number of patients with engel 1 outcomes both in the text and the table to reflect the 12 patients who were seizure free of the 19 with available outcomes.

5. The “outcome” field in the participants table contains decimal values (e.g., 1.1, 2.1) which are not explained by the dataset descriptor, which defines integer ILAE categories. If these represent Engel subscores (e.g., Ia = 1.1), this should be stated explicitly, as Engel and ILAE are distinct scales. Additionally, the “follow_up” field has no descriptor, though I infer it is the follow-up duration after treatment in years. If so, this is inconsistent with the text claiming this was assessed at 24 months post-surgery, as several entries are less than 2.

We have corrected the issues associated with our reporting of outcomes in our dataset. We have clarified in the table that we are referring to engel scores. We have also corrected our data records description of outcomes to describe the Engel outcomes scores and device implantations, and added in a description of the follow_up field to make the description consistent with the dataset.